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Enterprise Infrastructure Portfolio Case Study

Balcom Insurance Homelab | INFRA-BASE-v1

Proxmox VE	OPNsense	Active Directory	Windows Server	Wazuh	Network Segmentation

PROJECT GOAL

Build a stable, segmented, enterprise-style infrastructure baseline before adding SOC telemetry, detection engineering, incident-response exercises, and threat-hunting workflows.

8	4	64 GB	2 TB
Core VMs	Department VLANs	Host RAM	Dedicated backups

Fictional enterprise environment built for hands-on cybersecurity, networking, and systems-administration practice.

PROJECT OVERVIEW

What I Built

INFRA-BASE-v1 is the validated infrastructure layer of a simulated enterprise environment. I designed the network, deployed core Windows and Linux services, hardened inter-network access, tested authentication and file access, and built a repeatable recovery baseline before moving into SOC telemetry and detection work.

The environment includes Proxmox virtualization, OPNsense routing and segmentation, an Active Directory domain, four department workstations, a Windows file server, a Wazuh monitoring server, and a malware-analysis bridge with no routed path.

Why This Project Matters

- Security monitoring is only as useful as the infrastructure underneath it. This project forced me to understand routing, identity, DNS, policy, permissions, and recovery before relying on a SIEM.
- I validated both allowed and denied traffic paths instead of assuming firewall rules worked.
- Several real configuration failures occurred during the build. I diagnosed and corrected them rather than rebuilding around the problem.
- The final baseline gives future SOC projects a known-good starting point for controlled attack simulation, detection engineering, incident response, and threat hunting.

Skills Demonstrated

VLANs	Firewall Policy	AD DS	DNS/ DHCP	GPO	SMB/ NTFS	AGDLP	Linux	Wazuh	PowerShell	Troubleshooting	Backup & Recovery

Baseline Outcome

> **VALIDATED INFRASTRUCTURE BASELINE**
>
> Domain health, DNS, Group Policy, SMB access, Wazuh reachability, Internet access, segmentation controls, and post-reboot recovery were tested successfully. A complete offline VZDump backup set was also written to a dedicated external target.

ARCHITECTURE

Logical Design

OPNsense is the sole Layer-3 gateway for the routed lab networks. Proxmox internal bridges provide Layer-2 connectivity, while department networks are separated with VLANs. The malware bridge is intentionally isolated from OPNsense and has no physical uplink.

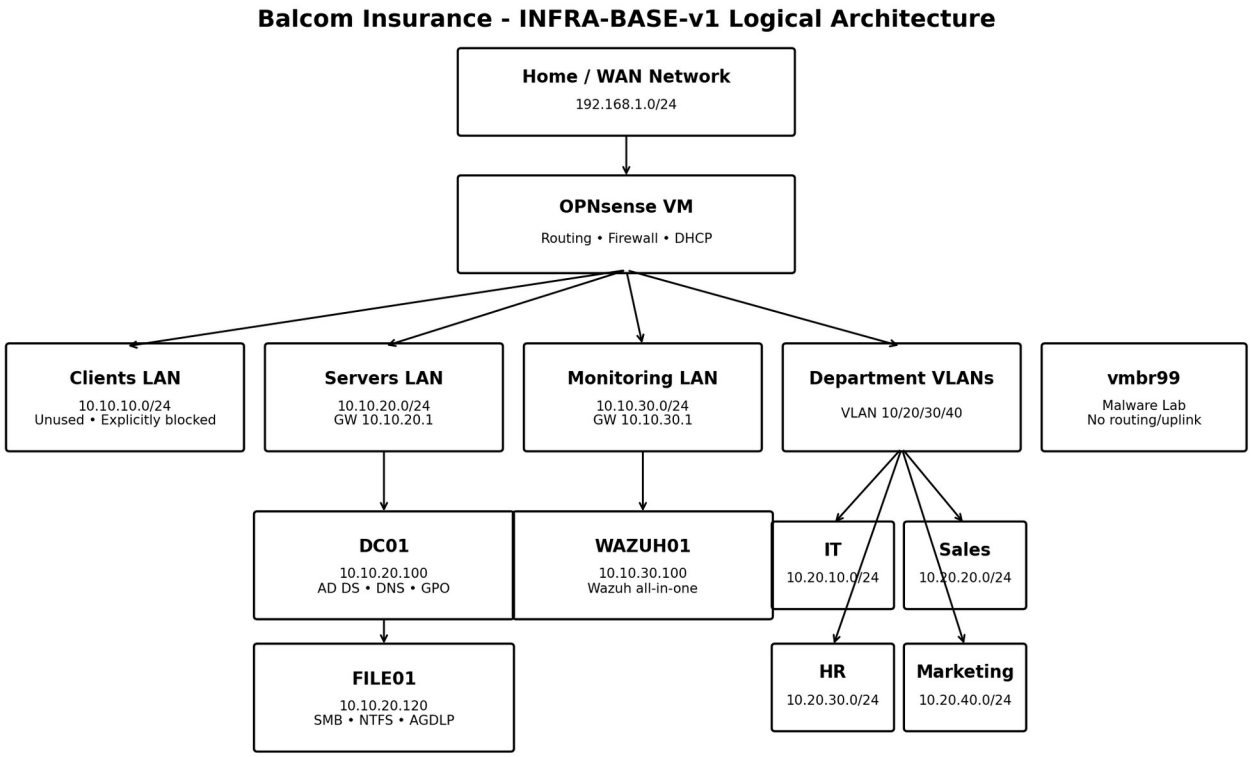


Figure 1. INFRA-BASE-v1 logical architecture.

KEY DESIGN DECISIONS

OPNsense owns Layer-3 routing. Proxmox internal bridges remain Layer 2. Department users are segmented by VLAN, infrastructure services live on dedicated server/monitoring networks, and the malware bridge is isolated from all routed paths.

INFRASTRUCTURE INVENTORY

Network & VLAN Inventory

Network	VLAN	Subnet	Gateway	Purpose
Home / WAN	-	192.168.1.0/24	192.168.1.1	Home uplink / Proxmox management
Clients LAN	-	10.10.10.0/24	10.10.10.1	Unused; explicitly blocked
Servers LAN	-	10.10.20.0/24	10.10.20.1	DC01 + FILE01
Monitoring LAN	-	10.10.30.0/24	10.10.30.1	WAZUH01
IT	10	10.20.10.0/24	10.20.10.1	IT workstation
Sales	20	10.20.20.0/24	10.20.20.1	Sales workstation
HR	30	10.20.30.0/24	10.20.30.1	HR workstation
Marketing	40	10.20.40.0/24	10.20.40.1	Marketing workstation
Malware Lab	-	Isolated L2 only	None	vmbr99; no routed path

Virtual Machine & Service Inventory

VM	Role	Addressing	RAM	Primary use
OPNsense	Firewall / router	WAN + internal	4 GB	Routing, firewall, DHCP
DC01	Windows Server / DC	10.10.20.100	6 GB	AD DS, DNS, GPO
FILE01	Windows member server	10.10.20.120	4 GB	SMB, NTFS, shares
WAZUH01	Ubuntu 24.04	10.10.30.100	12 GB	Wazuh 4.14.7 all-in-one
IT-PC01	Windows client	DHCP - VLAN 10	4 GB	IT workstation
SALES-PC01	Windows client	DHCP - VLAN 20	4 GB	Sales workstation
HR-PC01	Windows client	DHCP - VLAN 30	4 GB	HR workstation
MKT-PC01	Windows client	DHCP - VLAN 40	4 GB	Marketing workstation

When all listed VMs are powered on, the allocation is approximately 42 GB of the 64 GB host, leaving capacity for Proxmox overhead, cache, and future SOC workloads.

NETWORK SECURITY

OPNsense Segmentation Strategy

Firewall policy follows a service-first pattern: allow the exact internal services a zone needs, block access to other private networks, then permit Internet egress. OPNsense first-match processing makes rule order part of the security design.

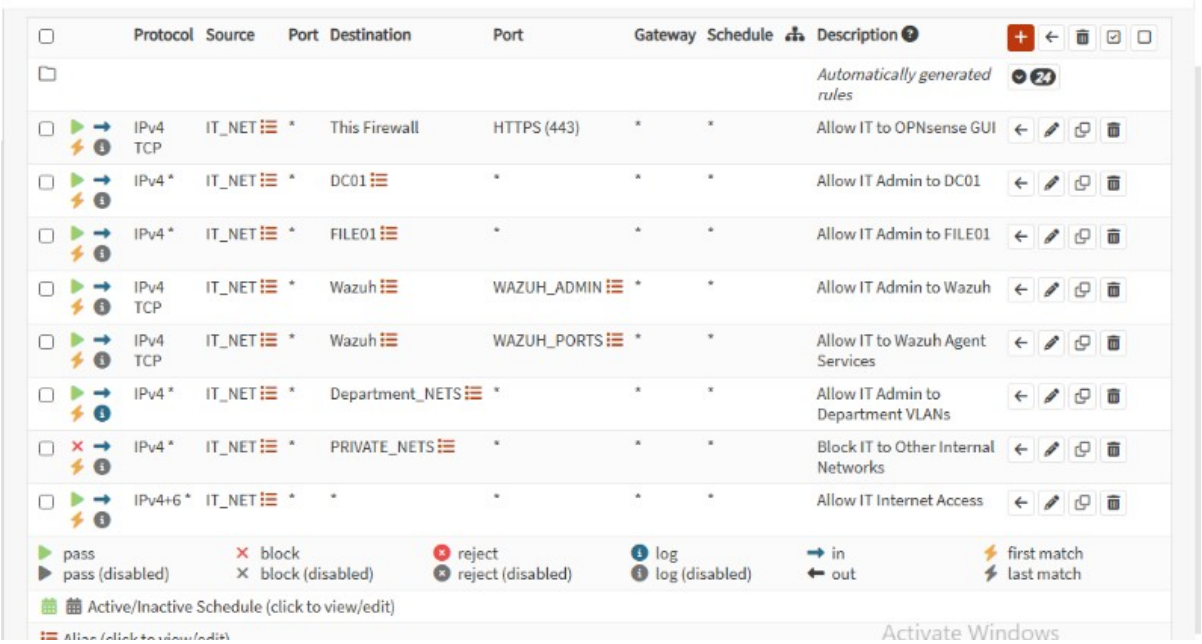


Figure 2. Final IT firewall policy: explicit management/service allows, broad private-network deny, then Internet access.

DESIGN PRINCIPLE

Administrative access is explicit. Department networks receive only required AD/DNS/RPC, file-service, Wazuh, and Internet paths. The unused Clients LAN is blocked, and vmbr99 has no routed path at all.

IDENTITY & POLICY

Active Directory Design

Domain: homelab.internal | NetBIOS: HOMELAB | Domain Controller: DC01

The Corporate OU structure separates Users, Workstations, Servers, Groups, and Service Accounts. Department-specific child OUs support targeted administration and policy application while keeping the directory easy to reason about.

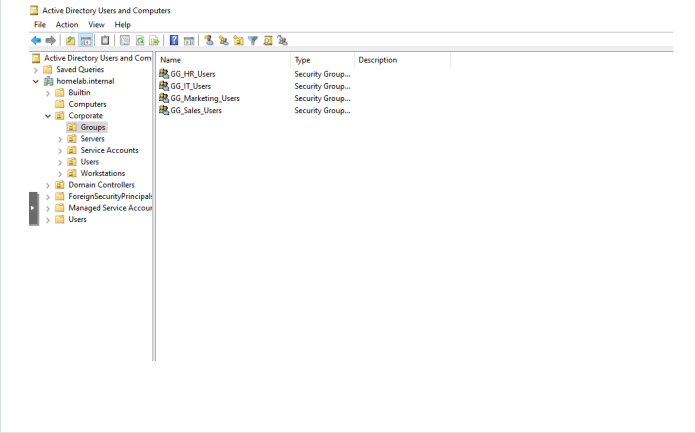


Figure 3A. Department security groups organized under Corporate\Groups.

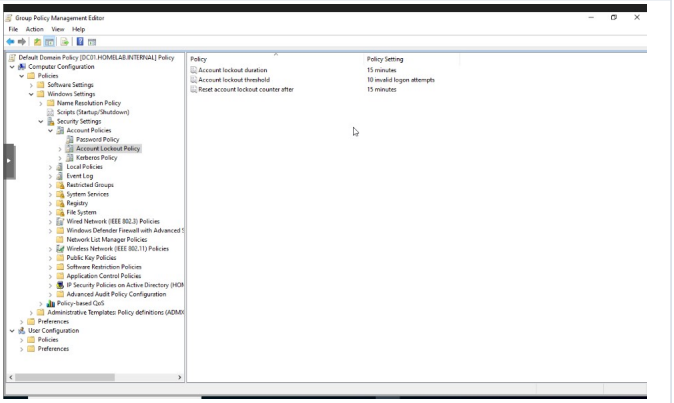


Figure 3B. Domain account lockout policy configured through Group Policy.

Group Policy Baseline

GPO	Scope	Validated baseline
GPO-Workstations-Baseline	Corporate\Workstations	Firewall enabled; Defender real-time protection; Guest disabled; Windows Update configured
GPO-Users-Baseline	User OUs	Baseline user policy validated with gpupdate / gpresult
Default Domain Policy	Domain	Complexity; 12-char minimum; history 12; lockout 10 attempts / 15 min

IDENTITY VALIDATION

Each Windows client was given a unique computer identity, joined to the correct department OU, and tested with Test-ComputerSecureChannel plus Group Policy refresh.

FILE01: File Services & Least Privilege

FILE01 is a domain-joined Windows Server with a separate 100 GB NTFS data disk. Department shares are stored under D:\Shares and use AGDLP-style group nesting so user membership and resource permissions remain separate concerns.

SMB share	NTFS path	AGDLP mapping
\\FILE01\IT	D:\Shares\IT	GG_IT_Users -> DL_IT_Share_RW
\\FILE01\Sales	D:\Shares\Sales	GG_Sales_Users -> DL_Sales_Share_RW
\\FILE01\HR	D:\Shares\HR	GG_HR_Users -> DL_HR_Share_RW
\\FILE01\Marketing	D:\Shares\Marketing	GG_Marketing_Users -> DL_Marketing_Share_RW

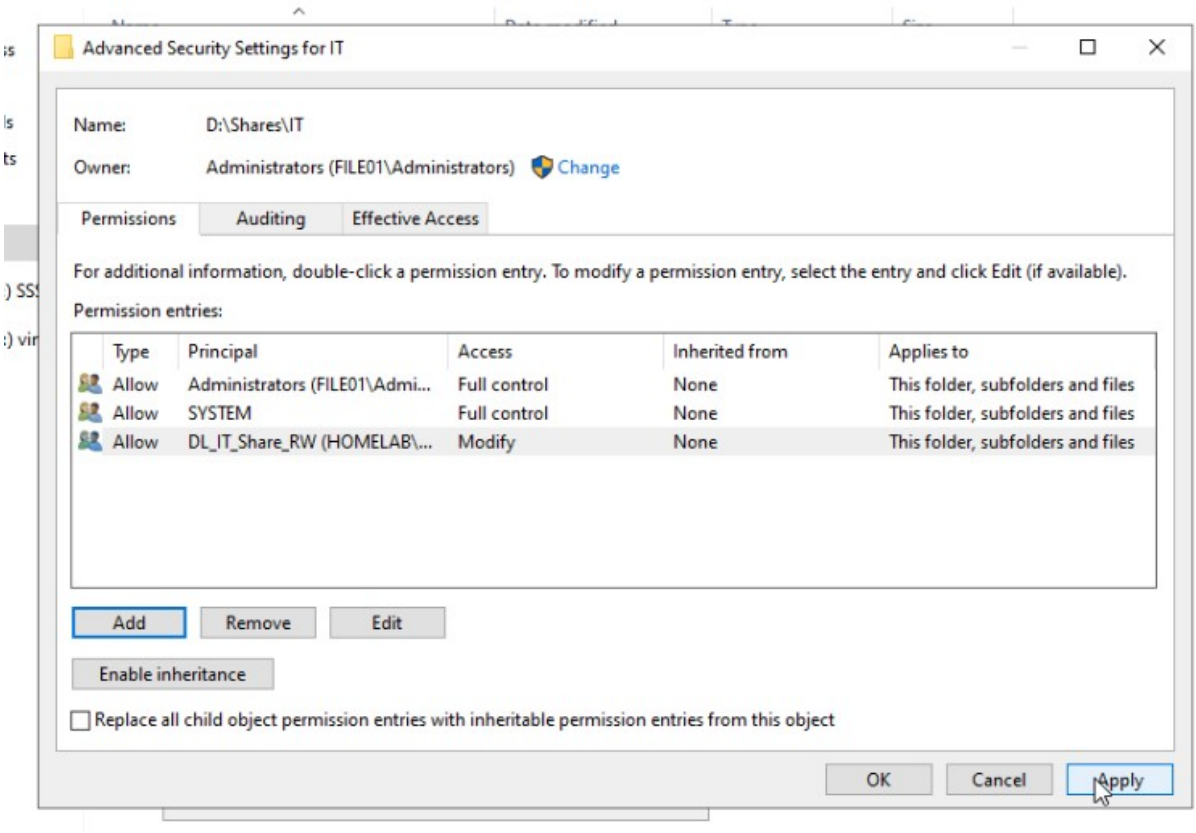


Figure 4. Final IT share permissions: Administrators and SYSTEM retain Full Control; the department Domain Local group receives Modify.

Share-level access is intentionally broad while NTFS acts as the effective security boundary. Positive and negative tests confirmed that matching department users could access their shares and cross-department access was denied.

SECURITY MONITORING

Wazuh Placement & SOC Readiness

WAZUH01 runs Ubuntu 24.04 with Wazuh 4.14.7 all-in-one on the dedicated Monitoring LAN at 10.10.30.100. The VM has 12 GB RAM. During deployment, storage pressure exposed that Ubuntu was not using the full virtual disk; I expanded the root LVM before completing the installation.

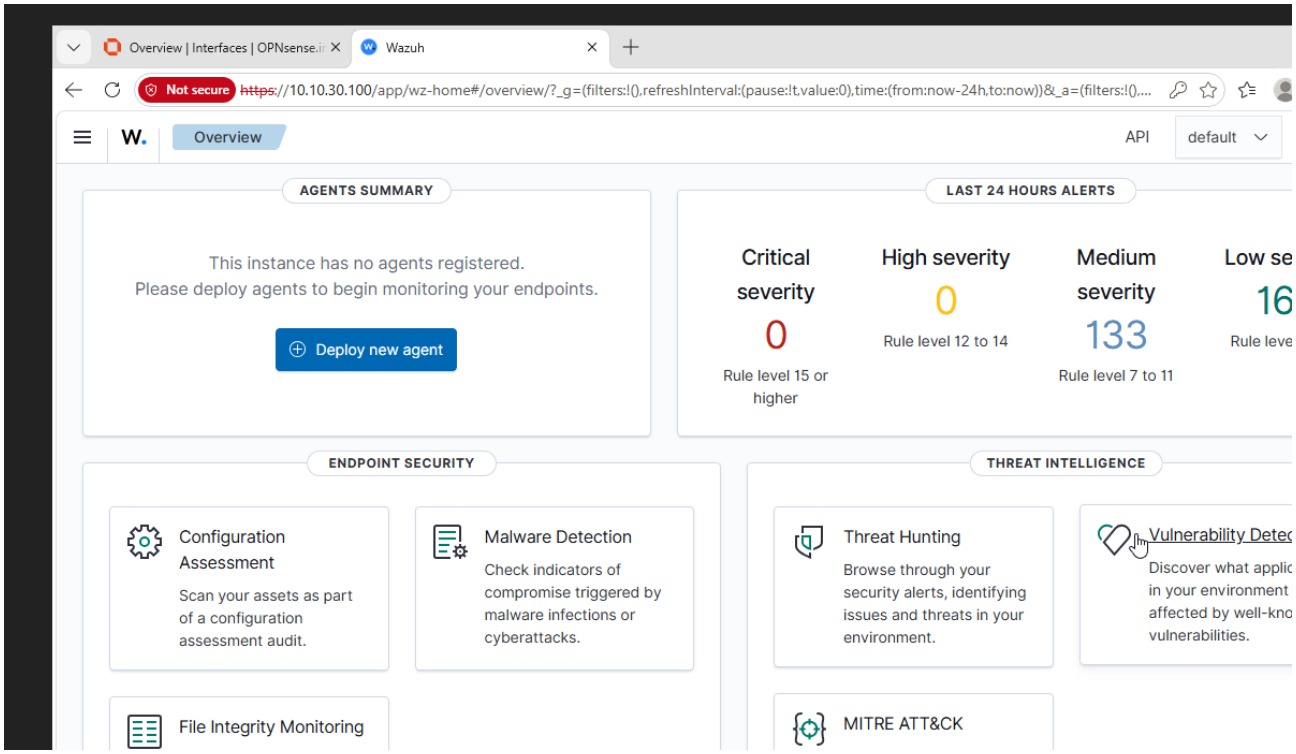


Figure 5. Wazuh dashboard deployed and reachable from the IT administration path. Agent enrollment intentionally follows the infrastructure freeze point.

SEGMENTATION

Clients and servers can reach required Wazuh service ports. The Monitoring LAN is prevented from initiating broad connections into corporate private networks, reducing unnecessary trust from the monitoring tier.

What Comes Next

- Deploy Wazuh agents to Windows endpoints and servers.
- Add Sysmon and centralized endpoint telemetry.
- Generate controlled attack activity and validate observable events.
- Write and tune detections, then document triage and incident-response workflows.
- Use the stable INFRA-BASE-v1 backup as the recovery point between experiments.

VALIDATION

Acceptance & Recovery Testing

The baseline was treated like a real change window: I tested the environment before declaring it stable, then performed a controlled shutdown, rebooted the Proxmox host, brought services back in dependency order, and repeated critical checks.

Control	Evidence / test	Result
AD health	dcdiag; dcdiag /test:dns	PASS
Domain trust	Test-ComputerSecureChannel on FILE01 and clients	PASS
Group Policy	gpupdate /force	PASS
DNS	Internal domain + Internet name resolution	PASS
File services	TCP 445 + department share access	PASS
Wazuh reachability	TCP 1514 from clients/servers; HTTPS 443 from IT	PASS
Segmentation	Cross-department, server-to-client, monitoring-to-corporate negative tests	PASS
Recovery	Controlled shutdown, host reboot, ordered startup, post-reboot validation	PASS

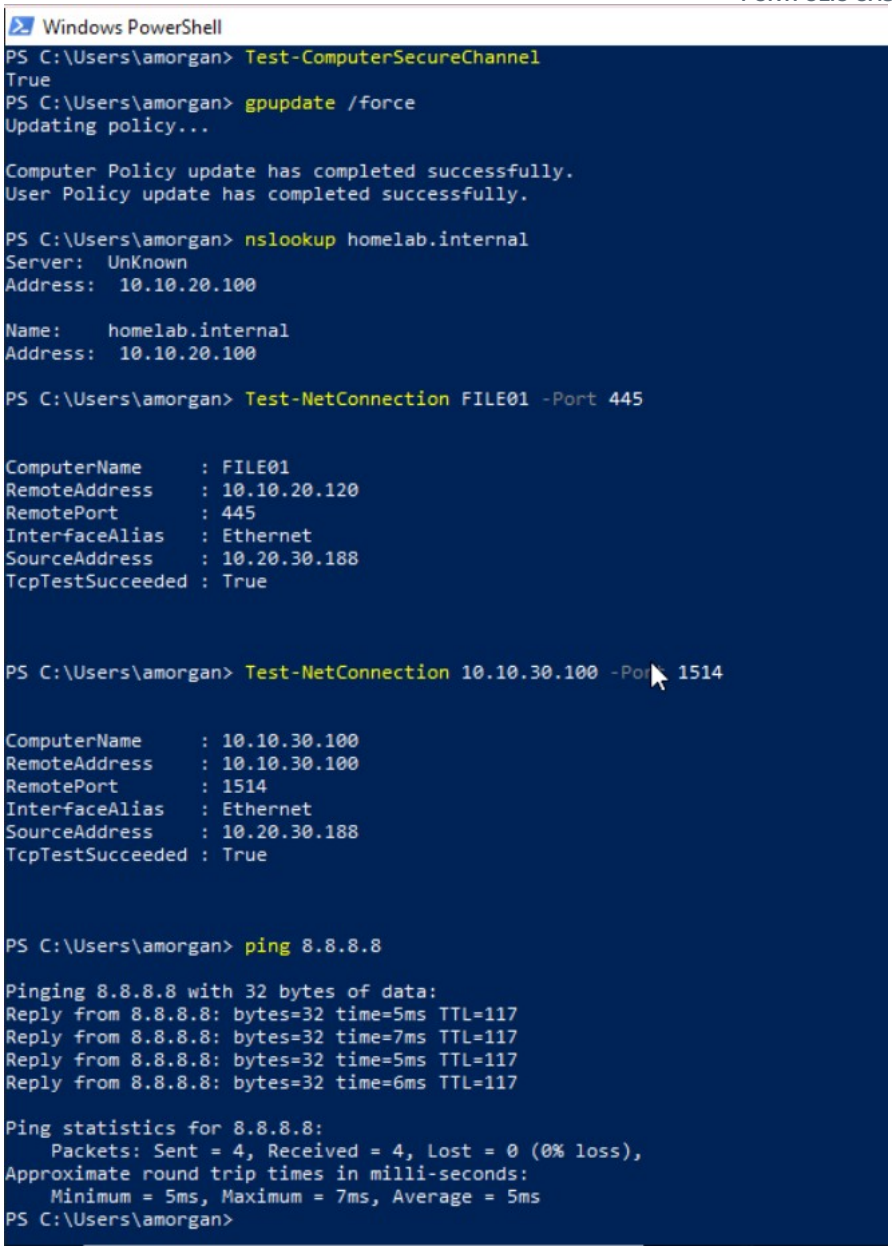


Figure 6. HR workstation acceptance test: secure channel, GPO refresh, internal DNS, FILE01 SMB, Wazuh 1514, and Internet connectivity all validated.

ENGINEERING LESSONS

Major Troubleshooting & Root-Cause Analysis

Duplicate gateway ownership

Proxmox vmbr1/vmbr2/vmbr3 carried .1 addresses that conflicted with OPNsense. I changed the bridges to Layer 2 only (inet manual), leaving OPNsense as the sole Layer-3 gateway.

Marketing VLAN parent

The Marketing VLAN was attached to the wrong OPNsense parent interface. Correcting the parent restored DHCP and stable connectivity.

Cloned Windows identity

Cloned clients shared a computer identity and caused machine-account trust failures. I repaired the secure channels, assigned unique hostnames, and documented Sysprep/generalization for future templates.

Wazuh storage sizing

Ubuntu had unused virtual-disk capacity outside the root logical volume. Extending the LVM filesystem resolved the installation failure.

AD RPC / GPO

Group Policy processing initially hung because the dynamic RPC firewall range was represented incorrectly. Correcting the rule restored immediate gpupdate processing.

OPNsense management lockout

A broad private-network block also matched the firewall management interface. I recovered through the console and added an explicit IT-to-This-Firewall HTTPS allow above the block.

Endpoint vs. network firewall

OPNsense permitted an IT-to-Sales ICMP test, but Windows Defender Firewall blocked the endpoint response. This demonstrated how layered controls can produce different observations at different enforcement points.

What I Learned

- A clean network design requires clear ownership of Layer-2 bridging versus Layer-3 gateway functions.
- Active Directory is highly dependent on DNS, RPC, stateful firewall behavior, and correct computer identity.
- Firewall ordering and aliases are operational security controls, not just configuration details.
- Recovery planning is part of engineering; a lab that cannot be restored safely is not a reliable security platform.
- Troubleshooting improved most when I validated one layer at a time instead of changing multiple variables at once.

RECOVERY & HANDOFF

INFRA-BASE-v1 Recovery Baseline

A dedicated 2 TB external disk was formatted as ext4 and added to Proxmox as a backup-only storage target. With all VMs shut down, I created offline ZSTD-compressed VZDump backups for every core VM. This gives the project a recovery artifact that is separate from the primary VM storage.

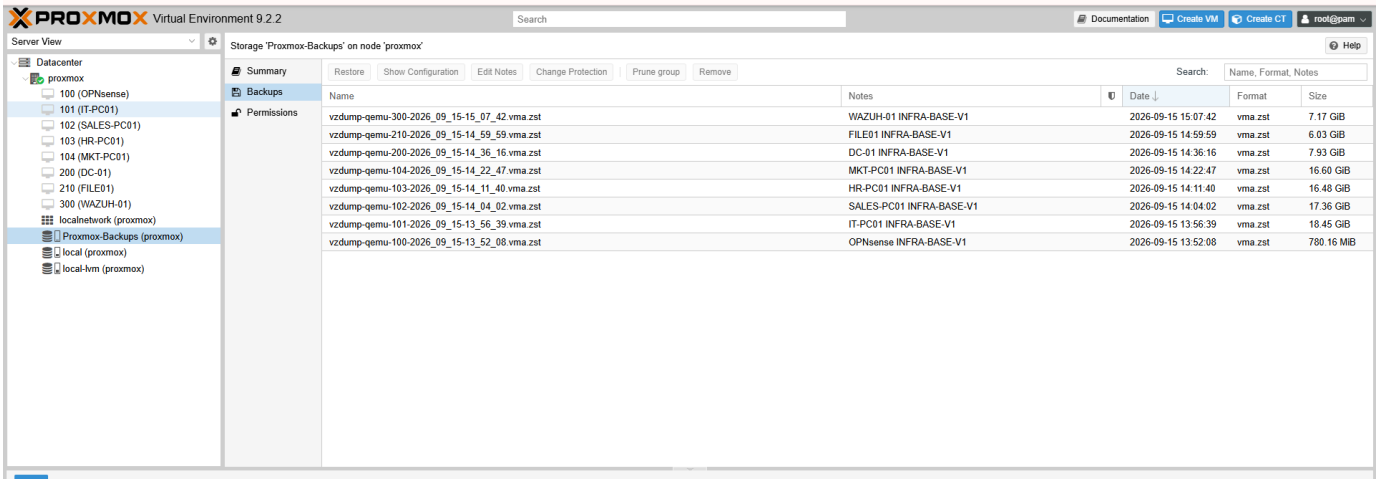


Figure 7. Complete INFRA-BASE-v1 VZDump backup set for all eight VMs on the dedicated Proxmox-Backups storage target.

RECOVERY MODEL

External VZDump backups are the authoritative recovery copy. Proxmox snapshots may be used as a convenient local rollback point, but they are not treated as a replacement for backups - especially for Active Directory.

Project Result

INFRA-BASE-v1 now provides a documented, segmented, recoverable enterprise-style foundation for the next security phase. Instead of building detections on an unstable lab, future SOC work starts from infrastructure whose routing, identity, access control, service dependencies, and recovery behavior have already been tested.

Next Portfolio Milestone

SOC-BASE-v1

Wazuh agents + Sysmon -> centralized telemetry -> detection engineering -> alert triage -> incident response -> threat hunting. Each phase will be tested against controlled activity and documented with evidence.

Built, validated, broken, fixed, and documented by Carlos Tiburcio.